Response to Reviewers — Submission 6222

Original title: From Colonial Spatial Structures to Sovereign Spatial Planning: Multimodal Network Modelling and Optimal Location of New Urban Hubs in the Democratic Republic of the Congo Using Spatial Data Science.

Revised title: Planning New Urban Hubs in Western Democratic Republic of the Congo through Conditional Location Optimization and an Assessment Framework for Functional Autonomy

Author: PAMBOLO MAYAMBO Néhémie. Consolidated revision with supplied-route audit: 12 October 2026.

Dear Editor and Reviewers, thank you for identifying gaps between the initial planning ambition and its evidence. The revision now includes a conditional new-urban-hub location module alongside the baseline accessibility audits. All 26 existing hubs are fixed, and screened new centres are optimized by additional modelled population coverage. Twenty-four executed runs compare two candidate grids, two population products, road classes and an optional water-proximity constraint. Functional autonomy is specified as an assessment framework, not a measured sovereignty score. The main objective uses no unelicited preference vector or invented expert response. Historical causal claims and construction endorsements remain withdrawn. We do not assert that all requirements are fulfilled: matched empirical household-access validation, current operational verification and functional feasibility remain outstanding. Historical electricity infrastructure and published national enterprise observations are now integrated into the manuscript and the consolidated package, with their epoch and scale limitations explicit.

1. Title, introduction, and the colonial-legacy claim

Reviewer comment: The reviewer requested either removal of the colonial claim from the title or a quantitative historical comparison.

Response: Addressed by removal and scope correction. The title now identifies conditional new-hub location optimization and a functional-autonomy assessment framework. Section I states that no colonial-era network is digitized, and Section III.F explains why current coverage cannot apportion fragmentation between colonial inheritance and post-independence investment. Historical motivation is no longer a model result. No invented diachronic comparison is supplied.

Location and evidence: Sections I, III.F and IV; revised title.

2. MST + three-neighbour topology and centrality

Reviewer comment: The reviewer requested sensitivity to alternative graph topologies and distance thresholds, including Kenge and Bandundu.

Response: New calculations compare MST + k for k = 1, 2, 3, 4, 5, and 8 and radius graphs at 15, 25, 40, 50, and 100 km. Kenge changes from betweenness rank 1 to 7 and Bandundu from 2 to 14 between k = 3 and 5. Radius 100 km yields 12 components and eight isolates. The centrality-based investment interpretation is withdrawn. The main model now uses mapped road geometry and preserves disconnection rather than forcing MST links. Closeness and reachability are exported for every node. Missing junction and restriction information remains a graph-quality limitation.

Location and evidence: Sections II.C and II.F, III.B and III.D; Fig. 3; Table VII; topology_node_metrics.csv and topology_summary.csv.

3. Sovereign hubs and functional polycentric planning

Reviewer comment: The reviewer asked how selected nodes can act as sovereign hubs and break dependency on Kinshasa.

Response: New Sections II.L–M and III.L–M restore the new-hub planning question while distinguishing conditional location optimality from realized functional autonomy. The model selects new grid centres in addition to all 26 existing hubs, with population coverage as the objective and environmental/attachment conditions. Twenty-four new runs have zero solver gap. Functional autonomy is defined through productive, energy, service, employment, exchange and dependence indicators with explicit missing evidence; a 15-km water-proximity constraint tests one resource-access condition. None of the old Yanganga/Ebale scores is relabelled as demonstrated sovereignty. New selected centres are sectors for investigation, not viable construction parcels. Relative autonomy and reduced Kinshasa dependence remain hypotheses requiring functional data. The consolidated revision now adds Sections II.N and III.N with historical high-voltage proximity for all 38 distinct selected centres and Table XXII. In the five-centre example distances range from 8.1 to 278.8 km. Published WBES 2024 national enterprise indicators are documented with their sampling universe. These additions do not establish local reliable power, economic independence or observed service coverage.

Location and evidence: Sections II.L–N and III.L–N; Fig. 6; Tables XIX–XXII; all_new_hub_scenarios.csv, all_new_hub_selections.csv and functional_evidence/results/.

4. Hydrological proxy specification

Reviewer comment: The reviewer requested the formula or GIS procedure behind high/moderate exposure.

Response: The original cached rule is recovered and specified: high if distance to mapped water ≤5 km and slope ≤2%; otherwise moderate if distance ≤15 km and slope ≤5%; otherwise low, with missing inputs unclassified. The five-query SRTM90 kilometre-scale

slope calculation is documented. We explicitly distinguish this heuristic from flood probability, depth, discharge, or a return-period map. The flag remains in the legacy audit and is not used to certify construction safety.

Location and evidence: Section II.F; candidate_weight_audit_metadata.json; legacy environment script.

5. Empirical validation using secondary observations

Reviewer comment: The reviewer requested validation against actual road, port, flight, field, or historical data.

Response: Partially addressed with executed external input and route checks, not validated regional population coverage. The satellite reference yields 20.93% length agreement within 100 m with tracks. We now also compare published 2018 Logistics Cluster road distances using the exact directed graph and both travel directions. Main-road Kinshasa–Matadi is 336.03 km versus 355 km published (−5.34%); Kinshasa–Bandundu is 922.62 km versus 490 km (+88.29%), and adding tracks still gives 879.89 km (+79.57%). Endpoint offsets below 0.014 km cannot explain the discrepancy. We report this unfavourable finding and withdraw any general validation claim. The catalogue's Matadi–Bandundu value violates the triangle inequality with the other two symmetric entries; it is not fitted as a shortest-path calibration target. The benchmark has unspecified terminals/itineraries and an eight-year epoch difference. The JICA and AICD observations remain catalogued with their incompatible targets and limited independence. No GPS, household-access observation or interview is invented. Empirical validation of the regional covering outcome remains outstanding. A follow-up now retrieves three actual OSM ways for Bac Bandundu and adds them using exact shared coordinates, without synthetic gap links. The conditional main-road path falls to 398.99 km, compared with the geographically specified 405-km Mongata route in the Logistics Cluster April 2024 report. This supports a structural explanation of the detour, conditional on ferry usability; current operation and household access remain unvalidated (Sections II.K and III.K, Fig. 5). Sections II.O and III.O now add the supplied 40-route audit. We executed four unchanged graph variants, yielding 160 recorded comparisons; main roads give 29 finite paths, 11 unavailable paths and mean/median absolute relative disagreement of 24.72%/3.30% against OSRM. This is a shared-OSM routing comparison, not independent field validation; OSRM minimizes driving time rather than our length objective. Two commercial distances are verified on the cited page, but their original measurement method is unknown. Matadi–Boma differs by 90.01% and Kinshasa–Mbanza-Ngungu by 8.04% from those advertised lengths. The unverified Scribd reproduction is excluded from confirmed documentary checks. We retain unfavourable discrepancies and missing paths rather than claiming a validated regional outcome. Tables XXIII–XXIV, references [46]–[47] and route_comparison_40/results/ provide the evidence.

Location and evidence: Sections II.G and II.J–K, III.I–K and IV; Figs. 5; Tables XIV–XVII; references [32]–[39]; route and satellite-road diagnostic CSVs. New audit: Sections II.O and III.O; Tables XXIII–XXIV; all supplied route inputs and per-scenario CSVs.

6. Joint uncertainty rather than weights alone

Reviewer comment: The reviewer requested simultaneous uncertainty in population, OSM completeness, speeds, and coordinates.

Response: Partially addressed by new joint calculations: two population products, one- and five-kilometre demand grids, main roads versus inclusion of tracks, connector limits 1/2.5/5 km, fixed 1% road-feature removal, and fixed 250-m hub-coordinate displacement. The full design has 96 configurations per p and 288 exact runs. For p = 10, coverage spans 40.40–63.35%. This envelope is not a confidence interval, and removal/displacement levels are uncalibrated stress assumptions. We withdrew travel-time claims, so no assumed speeds enter the revised objective. Road additions and distributions of positional or operational error are not calibrated. The extension adds 108 exact runs: an unperturbed baseline and five independent combined deletion/displacement realizations, both products, three hub counts, and 50/100/150-km coverage thresholds, at 5-km demand resolution. These tests remain uncalibrated scenario stresses, not confidence intervals. Section II.I now adds exact functional ANOVA of the complete categorical factorial design, using explicitly independent uniform mass on levels. Table XII reports first-order and total-effect indices; these are finite-design variance shares, not calibrated error probabilities. The extension adds province and detection-period comparisons plus 1,000 spatial-block resamples for each of two road sets and two block sizes. Track-inclusive agreement is 31.22% for 2019–2022 roads and 10.19% for 2023–2025. The 25-km resampling range is 12.81–30.95%, changing to 12.06–32.26% at 50 km. These are conditional reference-road ranges, not calibrated total-error or hub-coverage confidence intervals. No new covering optimization is counted. Section II.J and Table XVI explain the remaining sources of uncalibrated error. Twelve further exact optimizations test this specific ferry addition across both products, both road-class sets and p = 5/10/15, retaining one-kilometre demand and the 100-km threshold. Coverage is unchanged for p = 5 and 10 and rises by 0.049–0.079 percentage points for p = 15. This is a targeted mode-availability diagnostic, not a calibrated stochastic completeness distribution.

Location and evidence: Sections II.E, II.H–K and II.L, III.C and III.G–L; Tables II, VI, IX, XII, XVI–XVII and XIX–XX; joint_maximal_coverage_results.csv, repeated_joint_stress.csv and all_new_hub_scenarios.csv.

7. Justification of multicriteria weights

Reviewer comment: The reviewer requested empirical justification or expert elicitation for the criterion weights.

Response: The manuscript now labels the weights as assumptions, not expert elicitation or AHP. The audit distinguishes the six preliminary weights from the actual seven-variable environmental Monte Carlo vector. Repeating 5,000 draws around uniform weights changes dominant winners in three provinces. Crucially, Yanganga is the only compatible cached candidate in Tshuapa, making its 100% selection frequency automatic. These findings remove the previous robustness endorsement. No unperformed consultation is claimed; greenfield recommendations are withdrawn. A six-criterion 100-point elicitation form, face-to-face consultation guide and

executable response-analysis script are supplied. Actual responses remain absent. The script produces no expert weight or ranking from an empty form; with genuine responses it reports panel dispersion, mean/median aggregation, ties and leave-one-out sensitivity. The six-criterion form applies only to the supplementary grid screen, not the old seven-variable slope audit or the principal unweighted covering objective. This preparation does not fulfil expert elicitation itself. We have additionally executed a CRITIC-style dispersion/correlation weighting of the six admitted grid-score columns, separately at 10 and 25 km. Equations and scale conventions are explicit. At 10 km, seven of ten provincial top centres change versus the assumed weights; at 25 km four of nine change. Table XVIII reports the derived vectors. This provides a transparent matrix-derived comparator, not retrospective expert support for the original vector. Expert elicitation remains unperformed. The new-centre allocation in Section II.L uses no multicriteria preference weights: it maximizes additional population coverage and compares a separate water-access condition. Thus expert elicitation is not claimed or required for interpreting this explicitly unweighted location objective. The old weighted screens remain audits, not endorsements. The objective, service-distance limit and resource constraint remain policy/model choices that need substantive justification and external validation.

Location and evidence: Sections II.F, II.J–L and III.E, III.K–L; Tables VIII and XVIII; criterion audit outputs. The principal new-centre objective contains no preference-weight vector.

8. Population alternatives, road completeness, thresholds, grid resolution, forest definition, and Sobol analysis

Reviewer comment: The reviewer specified WorldPop versus alternatives, segment perturbations, thresholds 15/25/40/100 km, grid resolution, forest definition, and possible Sobol decomposition.

Response: The population alternatives and radius topology tests are completed as previously described. The extension now also compares complete 10- and 25-km candidate screens on a shared boundary mask with a common zero origin and recomputed complete-grid population potential. Native WorldCover samples compare strict tree exclusion with an extreme tree-admitted diagnostic. The official WDPCA protected-area view for October 2026 is applied with centre and full-cell intersection rules. The two grids yield 6,519 and 1,047 access/remoteness-eligible centres, with 1,298 and 268 full-cell protected-polygon intersections before land-cover screening. These calculations address the previously missing grid and hard-forest-rule sensitivity. They do not distinguish primary from degraded forest, certify tenure or flood safety, or calibrate a continuous uncertainty distribution. Section II.I now supplies exact categorical functional ANOVA for all six factorial inputs. For $p = 10$, the population-product first-order share is 86.07%, with 2.81% assigned to interactions. This finite-design result is not a Monte Carlo Sobol estimate. The strict full-cell screen retains 1,583 ten-kilometre candidates and 241 twenty-five-kilometre candidates. Équateur has no retained candidate on the latter grid. Among nine common provinces, the median top-centre shift is 212.22 km under reference weights and 52.02 km under uniform weights, demonstrating substantial grid sensitivity. Road additions and calibrated uncertainty distributions remain untested.

Location and evidence: Sections II.B, II.E–F, II.H–I and II.L, III.G–H and III.L; Tables II, IX–XII and XIX–XX; complete supplementary scenario and candidate-screen tables.

9. Complete population demand, standard urban definitions, and falsifiable elbow

Reviewer comment: The reviewer requested population cells or the full locality set, standard urban definitions, and a falsifiable hypothesis.

Response: The principal model now includes every positive selected population cell: 219,279 WorldPop one-kilometre cells (24,553 five-kilometre cells) and 147,956 GHS-POP one-kilometre cells (21,120 five-kilometre cells). Rural and disconnected demand remain in the denominator. Section II.I now replaces point-only interpretation with four-connected polygons from native GHS-SMOD codes 30, 23, and 22, labelled separately across the country mask before study-region selection. It identifies 47, 182, and 139 morphological components. Aligned native GHS-POP cells determine component population; 25/26 hub points lie in a component. Eight-connectivity is checked explicitly. These footprints are derived morphological components, not official UCDB identifiers or functional urban areas (Section III.H, Table XIII). The old p-median is retained only as an audit, solved exactly for $p = 2–20$. A specified three-successive-relative-gains-below-5% rule identifies no elbow; the earlier ten-hub claim is withdrawn. The regional model changes to maximal covering because the mapped-road graph is disconnected.

Location and evidence: Sections II.B, II.D, II.F and II.I–L, III.A, III.C–D and III.H, III.L; Tables III, V, XIII and XIX–XX; Figs. 2 and 4; elbow_rule_check.json.

Reproducibility and release status

The accompanying revision package contains the new Python scripts, verified derived tables, metadata, and figures. The existing Zenodo v1.0.0 archive (10.5281/zenodo.22287394) predates the revision and must not be represented as containing these new analyses. The original archive is retained unchanged. A new repository release and corresponding archive are needed to freeze the revised analysis publicly. The author should confirm the final release citation after depositing the supplied revision package. The current package includes the supplied cartographic/documentary files, original graph-function source hash, verified road-archive hash, 160 route comparisons, missing-path status and reproduction instructions. No synthetic distance from the earlier SIM-labelled file is used.

Summary of remaining substantive limitations

External road and published-route checks have been executed. The conditional ferry scenario explains a major structural component of the Kinshasa–Bandundu detour; current usability remains unverified. Spatial-block resampling strengthens characterization of the satellite reference but does not calibrate a joint model-error distribution. Compatible regional access observations and actual expert responses remain absent. Functional sovereignty, seasonal operations, forest integrity, tenure and constructability are not validated. The 100-km threshold is assumed. These outstanding requirements are substantive and the revision does not assert that all reviewer requests are satisfied. The new forty-route comparison and two commercial checks remain limited audits: field distances, true service access and calibrated error distributions are still absent. The finite-sample threshold agreement does not remove these limitations.